

Notes: Gehring & Schaudt (QJE, 2026)
Insuring Peace: Index-Based Livestock Insurance, Droughts, and Conflict

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1 Big picture

1.1 Central question and contribution

- **Main question.** Can a market-based insurance product reduce violent conflict caused by climate shocks? The paper studies whether Index-Based Livestock Insurance (IBLI) mitigates drought-induced conflict in Kenya.
- **Why this question matters.** Droughts intensify competition over scarce grazing land and water. In pastoralist economies, droughts push herders into longer-distance migration, often into areas occupied by farmers, urban settlements, protected areas, or other land users. Those encounters can escalate into conflict.
- **Why IBLI is different from standard social protection.** Unlike ordinary indemnity insurance, IBLI uses remote sensing and satellite-based drought indexes. Payouts are automated and do not require individual loss verification. This makes the product scalable and less vulnerable to moral hazard, manipulation, or conflict-induced administrative delay.
- **Main empirical result.** A one-standard-deviation increase in neighborhood IBLI coverage reduces the drought-conflict semi-elasticity by about 25% in the baseline specification.
- **Broad contribution.** The paper links climate finance, development, and conflict economics. It shows that insurance can be more than a welfare-smoothing tool: it can function as conflict-prevention infrastructure.

1.2 The causal chain in the paper

The paper's logic can be summarized as:

Drought $\Rightarrow$ forage/water scarcity $\Rightarrow$ distress sales and migration $\Rightarrow$ encounters over contested resources $\Rightarrow$ conflict.

IBLI weakens this chain through:

IBLI coverage $\Rightarrow$ early payout / asset protection $\Rightarrow$ smaller herds, better liquidity
 $\Rightarrow$ price stabilization $\Rightarrow$ less migration pressure $\Rightarrow$ less conflict.

1.3 Why the paper emphasizes neighborhood exposure

- Conflict often occurs *away from* the drought-hit pastoralist home area, because herders migrate to find grazing resources.
- The relevant treatment is therefore not only whether the outcome cell itself is covered by IBLI. It is whether surrounding pastoralist areas that could send migrants into the cell are covered.
- The paper builds inverse-distance weighted neighborhood measures of drought and IBLI coverage to capture this spatial spillover logic.

Detailed interpretation: This is one of the paper's most important design choices. If the authors measured only own-cell drought and own-cell insurance, they would miss the main mechanism: drought-origin shocks generating conflict in destination cells.

1.4 Key mechanisms

- **Rapacity / resource-competition channel.** Drought raises the value of scarce grazing resources and water. Larger herds intensify pressure on common-pool resources. IBLI can reduce precautionary herd holding and thereby reduce the intensity of resource competition.

- **Opportunity-cost channel.** Drought lowers income and forces distress sales, reducing the opportunity cost of conflict. IBLI cushions income and stabilizes livestock prices, making conflict less attractive.
- **Migration-pressure channel.** Insurance payouts reduce the need for long-distance migration, lowering the probability of conflictual encounters with other land users.
- **General-equilibrium price channel.** When many pastoralists sell livestock during droughts, prices fall. IBLI can reduce aggregate distress sales, stabilizing cattle prices even for some households not directly insured.

1.5 What should be remembered after reading the paper

- The main outcome is the probability of conflict at a fine grid-cell level.
- The main explanatory variable is an interaction between neighborhood drought severity and neighborhood IBLI coverage.
- The key coefficient is negative: IBLI weakens the drought-conflict relationship.
- Robustness exercises focus on rollout selection, other interventions, alternative drought and conflict measures, spatial definitions, East Africa comparisons, and within-rollout-cluster heterogeneity.
- Mechanism evidence supports smaller herds, stabilized prices, smoother income, and less long-distance migration.

2 Data and measurement

2.1 IBLI: institutional design and rollout

- IBLI was piloted in Marsabit in 2010 and gradually rolled out across eligible pastoralist regions.
- Eligible areas are traditional pastoralist regions, many of which are arid or semi-arid.
- Insurance units are geographic areas for which drought indexes are calculated.
- Private insurers sell policies, while the Kenyan Livestock Insurance Program (KLIP) provides subsidized coverage for vulnerable pastoralists.
- IBLI coverage can enter through private purchases, subsidized coverage, or program availability in an insurance unit.
- Payouts are triggered by remote-sensing drought measures, not by individual claims adjustment.

Detailed interpretation: IBLI’s automatic index-based design makes it credible in an empirical conflict setting. If payouts were manually determined after conflict, payout timing and amounts might be endogenous to the conflict process. The satellite-index design reduces that concern.

2.2 What makes IBLI an asset-protection product

- A standard insurance contract often pays after losses are realized.
- IBLI pays when drought conditions cross a threshold, before all livestock losses occur.
- Pastoralists can use payouts to buy feed, water, veterinary care, or other inputs that protect livestock.
- Therefore, IBLI can reduce the need for migration and distress sales before they become severe.

Detailed interpretation: This timing distinction is essential. If the payout arrived only after herds were lost, it would smooth consumption but might not prevent migration-driven conflict. Preemptive payouts help preserve both household assets and local peace.

2.3 Conflict data

- The main conflict outcome comes from ACLED.
- The unit of observation is a $0.1^\circ \times 0.1^\circ$ grid cell by period.
- The main outcome is an indicator for whether at least one conflict event occurs in the cell-period.
- Robustness checks use alternative conflict outcomes, including subsets by actor type, event type, geographical precision, and pastoralist involvement.
- The paper also considers intensive-margin outcomes in robustness checks.

Detailed interpretation: A binary conflict indicator is useful because conflict-event counts can be noisy and highly skewed. The indicator asks whether drought and IBLI change the incidence of conflict in a cell-period.

2.4 Drought measurement

- The baseline drought proxy is the rain deficit, defined as log rainfall multiplied by negative one.
- A higher rain deficit means lower rainfall and therefore more severe drought.
- Rainfall is attractive because it is plausibly exogenous to conflict and policy rollout.
- Robustness checks use alternative drought measures such as aridity and dry matter productivity.

Detailed interpretation: Rainfall is not a perfect measure of forage availability, but it is less likely to be contaminated by conflict or economic activity than downstream measures. This explains why the baseline uses rainfall and robustness checks use more livestock-specific proxies.

2.5 Neighborhood variables

Let j index cells around outcome cell i . The paper constructs neighborhood exposure variables using inverse-distance weights:

$$RainDef_{it}^N = \sum_{j \neq i} w_{ij} RainDef_{jt}, \quad IBLI_{it}^N = \sum_{j \neq i} w_{ij} IBLI_{jt}.$$

Weights decay with distance, so nearby cells matter more than distant cells.

- Neighborhood drought captures drought conditions in areas that may send pastoralist migrants.
- Neighborhood IBLI captures whether potential origin areas have insurance coverage.
- The interaction captures whether insurance in drought-origin neighborhoods mitigates conflict in destination cells.

Detailed interpretation: The neighborhood structure embeds the pastoralist migration mechanism directly into the empirical design. It is not just a spatial-control device; it is how the model maps drought origins to conflict destinations.

2.6 Other interventions and controls

- The Hunger Safety Net Program (HSNP) is a major cash-transfer program in northern Kenya that may also smooth shocks.
- Development aid projects, especially agriculture-related aid, may reduce drought-conflict sensitivity.
- The paper controls for these interventions at cell and neighborhood levels.
- It also uses cell fixed effects, period fixed effects, and alternative spatial and temporal controls.

Detailed interpretation: If HSNP or aid programs were rolled out in the same places and times as IBLI, the estimated IBLI effect might actually reflect cash transfers or aid. The paper therefore treats overlapping interventions as a central omitted-variable threat.

2.7 Mechanism data

- **Herd size:** county-level livestock data from the Kenyan Ministry of Agriculture.
- **Livestock prices:** cattle prices from livestock markets.
- **Income shocks:** Afrobarometer outcomes, including hunger and no cash income.
- **Migration pressure:** distance from conflict events to ethnic homelands using ACLED actor information and Murdock homelands.

Detailed interpretation: The mechanism evidence is not a single test. It is a triangulation across multiple data sources that each capture a different link in the causal chain from drought to conflict.

3 Models and empirical specifications

3.1 Main conflict specification

The baseline equation is:

$$Conflict_{it} = \delta_1 RainDef_{it}^N + \delta_2 IBLI_{it}^N + \delta_3 (RainDef_{it}^N \times IBLI_{it}^N) + X'_{it}\xi + \eta_i + \gamma_t + \varepsilon_{it}.$$

- $Conflict_{it}$ is the conflict indicator in cell i and period t .
- $RainDef_{it}^N$ is neighborhood drought severity.
- $IBLI_{it}^N$ is neighborhood IBLI coverage.
- η_i are cell fixed effects.
- γ_t are period fixed effects.
- X_{it} are optional controls.

Detailed interpretation: δ_1 measures how drought increases conflict when insurance coverage is at its baseline standardized level. δ_3 measures how much insurance coverage changes that drought effect. The paper's central claim is that $\delta_3 < 0$.

3.2 Conflict mitigation percentage

The paper often interprets the interaction as a mitigation percentage:

$$Mitigation = \frac{\delta_3}{\delta_1}.$$

Because δ_3 is negative and δ_1 is positive, this ratio is a reduction in the drought-conflict effect.

Detailed interpretation: This statistic is easier to interpret than the raw interaction coefficient. It answers: by what share does a one-standard-deviation increase in IBLI coverage reduce the conflict effect of a one-standard-deviation drought increase?

3.3 Interpretation of fixed effects

- Cell fixed effects remove time-invariant local conflict risk, geography, ethnic composition, land-use history, and average program eligibility.
- Period fixed effects remove common shocks such as national politics, macroeconomic shocks, broad drought years, or changes in conflict reporting.
- Some robustness and heterogeneity specifications add more demanding spatial-time fixed effects, including rollout-cluster-period fixed effects.

Detailed interpretation: The identifying variation is not simply that insured areas are different from uninsured areas. It is how the drought-conflict relationship changes as insurance coverage changes across space and time.

3.4 Mechanism: herd-size specification

The herd-size mechanism can be summarized as:

$$HerdSize_{ct} = \beta IBLI_{ct} + \mu_c + \tau_t + u_{ct}.$$

- c indexes counties.
- $HerdSize$ can be total cattle, high-grade cattle, or beef cattle.
- A negative β indicates smaller herds after IBLI coverage.

Detailed interpretation: Smaller herds support the idea that insurance substitutes for livestock as precautionary savings. This weakens the rapacity/resource-pressure channel during droughts.

3.5 Mechanism: price-stabilization specification

$$\log(Price_{mt}) = \beta_1 RainDef_{mt}^N + \beta_2 IBLI_{mt}^N + \beta_3 (RainDef_{mt}^N \times IBLI_{mt}^N) + FE + u_{mt}.$$

- m indexes livestock markets.
- The outcome is the log cattle price.
- A positive interaction means IBLI softens drought-induced price declines.

Detailed interpretation: When drought pushes many households to sell at the same time, prices collapse. IBLI can reduce distress sales, stabilizing prices and raising the opportunity cost of violence.

3.6 Mechanism: income-smoothing specification

The Afrobarometer mechanism regressions use outcomes such as hunger and no cash income:

$$Outcome_{rt} = \beta_1 RainDef_{rt}^N + \beta_2 IBLI_{rt}^N + \beta_3 (RainDef_{rt}^N \times IBLI_{rt}^N) + FE + u_{rt}.$$

Detailed interpretation: The paper tests whether IBLI weakens the relationship between drought and adverse household welfare outcomes. Less hunger and fewer cash shortages make conflict less attractive and reduce desperate coping strategies.

3.7 Mechanism: migratory pressure specification

For conflict event k linked to homeland e in period t :

$$\log Distance_{ket} = \delta_1 RainDef_{et}^H + \delta_2 IBLI_{et}^H + \delta_3 (RainDef_{et}^H \times IBLI_{et}^H) + \eta_e + \gamma_t + u_{ket}.$$

- *Distance* is the distance from the conflict event to the relevant ethnic homeland.
- A positive drought coefficient means drought pushes conflicts farther from homelands.
- A negative interaction means IBLI reduces that drought-induced distance.

Detailed interpretation: If IBLI reduces long-distance migration pressure, drought-related conflicts should occur closer to homelands. This is a direct test of the migration mechanism.

3.8 Inference

- The paper uses spatially robust inference, including Conley-style standard errors in key specifications.
- Robustness checks vary the distance cutoff and clustering choices.
- Randomization-inference exercises test whether the observed interaction is unusually large relative to placebo rollouts.

Detailed interpretation: Spatial conflict data are correlated across nearby cells. The inference strategy is designed to avoid overstating precision when conflict and drought shocks spill across space.

4 Identification and empirical design

4.1 The core identification problem

The paper wants to know whether IBLI reduces drought-induced conflict. The problem is that IBLI was not randomly assigned. Eligible pastoralist areas received coverage in stages, and rollout timing could be correlated with conflict risk, drought risk, political priorities, or other social-protection programs.

Detailed interpretation: The design is quasi-experimental. The strongest source of exogeneity is rainfall variation, while the main vulnerability is rollout selection.

4.2 What provides identifying variation

- **Rainfall shocks:** drought severity varies across cells and periods in ways plausibly unrelated to contemporaneous conflict decisions.
- **Staggered rollout:** IBLI coverage arrives in different eligible areas at different times.
- **Spatial exposure:** neighborhood measures allow drought in pastoralist origin regions to affect conflict in destination regions.
- **Fixed effects:** cell and time fixed effects absorb many baseline differences and common shocks.

Detailed interpretation: Identification comes from comparing drought-conflict sensitivity before and after nearby IBLI coverage, while controlling for fixed local characteristics and period shocks.

4.3 Why the case study matters

The Samburu-Laikipia-Isiolo-Meru case study is not merely descriptive. It motivates the empirical design:

- droughts push pastoralists farther from home areas;
- conflicts cluster in contested land-use zones rather than always in drought-origin cells;
- migration paths and land-use boundaries imply spatial spillovers;
- eligible and ineligible areas coexist around migration destinations.

Detailed interpretation: The case study explains why the paper should estimate effects at the neighborhood level and why mixed land-use areas are expected to display stronger mitigation.

4.4 Rollout-selection checks

The paper examines whether pre-treatment drought-conflict elasticities predict:

- IBLI eligibility;
- timing of coverage among eligible units;
- conflict trends before rollout;
- rainfall trends that could indicate adaptive rollout to changing drought risk.

Detailed interpretation: These checks address the concern that IBLI was deliberately rolled out first to areas where drought-conflict risk was already improving or where conflict mitigation would have occurred anyway.

4.5 Other-program confounds

- HSNP cash transfers can also smooth income shocks.
- Development aid can affect irrigation, agriculture, conflict prevention, or local welfare.
- The paper controls for these programs directly and shows the IBLI interaction remains stable.

Detailed interpretation: This matters because a cash-transfer program with similar geographic coverage could create the same empirical pattern. Stability after controlling for HSNP and aid strengthens the interpretation that IBLI itself matters.

4.6 Robustness logic

The paper varies:

- drought proxies;
- conflict definitions;
- time periods;
- neighborhood distance decay functions;
- standard-error calculations;
- cell sizes;
- inclusion/exclusion of pre-rollout years;
- spillover controls from neighboring countries;
- East Africa sample restrictions.

Detailed interpretation: Robustness is important because both conflict measurement and spatial exposure construction involve many choices. The paper shows the main result is not driven by one arbitrary definition.

4.7 Within-rollout-cluster identification

One of the strongest checks uses within-rollout-cluster-period variation. The paper compares cells in the same rollout cluster and period that differ in expected exposure to pastoralist migration and contested land use.

Detailed interpretation: This design absorbs any unobserved shock or selection factor common to a rollout cluster in a given period. The remaining variation comes from where drought-migration conflict risk should be strongest within the same treated cluster.

4.8 What the design can and cannot prove

- The design can credibly show that conflict becomes less sensitive to drought when nearby pastoralist regions have IBLI coverage.
- It can show that the pattern survives extensive robustness and is strongest where the mechanism predicts.
- It cannot randomly assign coverage or directly observe every migration decision.
- It cannot prove that every insured household used payouts for feed or veterinary care.
- It cannot fully separate all general-equilibrium channels, such as price stabilization benefiting uninsured households.

Detailed interpretation: The empirical case is strongest because many pieces point in the same direction: baseline estimates, rollout checks, placebo tests, alternative samples, contested-land heterogeneity, and multiple mechanism tests.

5 Section-by-section study guide

5.1 Introduction

- Establishes drought-induced conflict as a major climate-security problem.
- Introduces IBLI as a scalable intervention using remote sensing and automated payouts.
- States the main estimate: a one-standard-deviation increase in neighborhood IBLI coverage mitigates the drought-conflict effect by about 25%.
- Previews the mechanisms: herd-size reduction, income smoothing, price stabilization, and less migration pressure.

How to read it. Focus on the causal chain and why the intervention is unusual: it is a private/market-based financial product with public conflict-reduction effects.

5.2 Setting

- Explains how IBLI works and why index-based payouts are feasible in remote pastoralist settings.
- Uses the central Kenya case study to show that conflict occurs where drought migration meets contested land use.
- Clarifies why own-cell treatment is insufficient and neighborhood treatment is needed.

How to read it. The setting section is the foundation for the empirical design. The maps are not decoration; they justify the spatial model.

5.3 Data

- Describes ACLED conflict data, rainfall/drought data, IBLI coverage and payout data, land-use and pastoralist measures, and mechanism datasets.
- Defines the $0.1^\circ \times 0.1^\circ$ grid-cell panel.
- Explains neighborhood construction and inverse-distance weighting.

How to read it. Track the difference between cell-level and neighborhood-level variables. The outcome is in the cell, but treatment exposure is mostly in the neighborhood.

5.4 Identification strategy

- Presents the baseline interaction model.
- Discusses rollout endogeneity and how the paper evaluates it.
- Introduces balance tests, placebo timing, robustness to other interventions, and spatial inference.

How to read it. The key question is whether IBLI coverage is as-good-as exogenous conditional on controls and fixed effects. The paper cannot randomize, so it builds credibility through many converging checks.

5.5 Main results

- Shows baseline drought-conflict estimates and the negative interaction with IBLI.
- Translates coefficients into a mitigation percentage.
- Tests alternative definitions and interventions.

How to read it. Pay attention to the interaction coefficient, not just the main effect of IBLI. IBLI is predicted to matter especially during droughts.

5.6 Mechanisms

- Tests whether IBLI reduces herd sizes.
- Tests whether IBLI stabilizes livestock prices during droughts.
- Tests whether IBLI smooths household welfare shocks.
- Tests whether IBLI reduces drought-induced conflict distance from homelands.

How to read it. The mechanism section does not rely on one dataset. It triangulates the pathway from financial insurance to reduced conflict.

5.7 Conclusion

- Interprets IBLI as a complement to state-building and political reforms.
- Discusses the DRIVE initiative and scaling to the Horn of Africa.
- Emphasizes the need for future data collection on take-up, coverage intensity, and large-scale program impacts.

How to read it. The conclusion turns a Kenya-specific quasi-experiment into a broader policy argument about climate adaptation, insurance, and conflict prevention.

6 Tables and figures

6.1 Figure I: Index-Based Livestock Insurance in Kenya

Read:

- The figure provides a process diagram for IBLI in Kenya.
- Insurance companies provide IBLI coverage in eligible units.
- Pastoralist households purchase coverage for tropical livestock units or receive subsidized coverage.
- Satellite monitoring measures drought conditions in the insurance unit.
- If drought thresholds are crossed, automated payouts are made.
- The figure also shows how contract periods correspond to rain and dry seasons.

Detailed interpretation:

- The central design feature is that payouts depend on an externally measured index, not on individual livestock losses.
- This mitigates moral hazard because households cannot easily manipulate satellite drought conditions.
- It reduces adverse selection because the contract is tied to area-level risk rather than privately known household losses.
- It also reduces endogeneity concerns in this paper because payouts are triggered by pre-specified remote-sensing rules rather than by conflict events or political discretion.

Economic message: IBLI is best interpreted as an asset-protection tool. Its value for conflict prevention comes from paying early enough to prevent distress migration and livestock liquidation, not merely from compensating households after losses occur.

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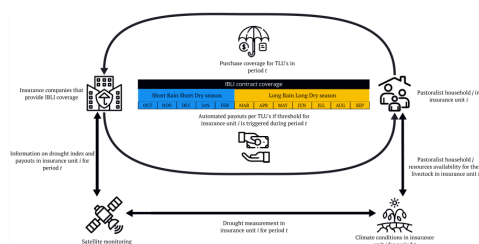


FIGURE I
Index-Based Livestock Insurance in Kenya

The figure provides a simplified scheme of the processes underlying IBLI. An insurance contract can also start coverage in the long rain long dry season, running from March to February.

(2004) provides a detailed summary of IBLI and discusses its impact.

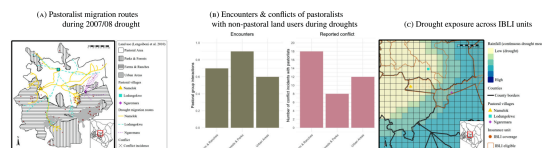


FIGURE II
Droughts, Conflicts, and IBLI in the Samburu-Laikipia-Isiolo-Meru Counties

Panel A shows the digitized map from Lengoboni, Bregt, and van der Molen (2010) with the different land-use types for the Samburu-Laikipia-Isiolo-Meru region and the drought migration routes and villages for the Nanyuki, the Ledungokwe, and the Ngarema pastoral groups. Panel B plots reported encounters and conflicts by land-use types in the study region based on 41 respondents interviewed by Lengoboni, Bregt, and van der Molen (2010). Conflict incidents are from ACLED (Raleigh, Kishi, and Linke 2023). Panel C shows counties and insurance units to illustrate the variation in IBLI coverage based on whether a county is eligible and on the timing of eligible counties receiving IBLI coverage. Samburu is the county in the north (IBLI coverage in 2017), Laikipia in the southwest (not eligible), Meru in the southeast (not eligible), and Isiolo in the east (IBLI coverage in 2013) (Fava et al. 2021). Rainfall is used to approximate the drought conditions in an insurance unit during the Short Rain Short Dry (SRSD) and Long Rain Long Dry (LRLD) seasons that make up our 12-month periods.

6.2 Figure II: Droughts, Conflicts, and IBLI in the Samburu-Laikipia-Isiolo-Meru Counties

Read:

- Panel A maps pastoralist home villages and migration routes in the case study region.
- Panel B combines survey encounters and ACLED conflict patterns.
- Panel C overlays IBLI eligibility, rollout timing, and hypothetical drought conditions.
- The figure shows that conflicts are concentrated in contested land-use areas rather than only in pastoralist homelands.
- It also shows that IBLI eligibility and coverage vary across nearby areas and over time.

Detailed interpretation:

- The figure motivates the core spatial logic of the paper. Drought-origin cells and conflict-destination cells differ.
- Laikipia and Meru are particularly important because they contain destination areas where pastoralists interact with farmers, towns, ranches, or protected lands.
- The case study explains why neighborhood variables should enter the regression and why contested land-use heterogeneity is a credible mechanism test.
- The visual evidence also helps distinguish pastoralist conflict from generic local violence: the patterns line up with drought migration routes.

Economic message: Drought-induced conflict is a migration-spillover problem. Insurance coverage in pastoralist origin areas can reduce conflict in neighboring destination areas.

6.3 Figure III: Droughts, IBLI Coverage, and Payouts + Figure IV: Neighborhood Definition and Distance Weights

Read:

- Figure III links drought severity, IBLI coverage, and payout-relevant areas.
- It shows that IBLI payouts are connected to drought conditions in insurance units.
- Figure IV illustrates how neighborhood exposure is calculated using distance-decay weights.
- Nearby cells receive larger weights; far-away cells receive lower weights.
- The figures together explain the construction of $RainDef^N$ and $IBLI^N$.

Detailed interpretation:

- Figure III validates the operational connection between drought conditions and insurance payouts.
- Figure IV is central to the empirical strategy because it shows why the treatment is not measured only inside the outcome cell.
- The distance-decay assumption corresponds to the idea that migration risk and conflict spillovers become weaker as distance increases.
- Alternative decay functions are later used as robustness checks, which is important because the exact spatial spillover function is not directly observed.

Economic message: These figures translate a qualitative migration story into measurable variables: drought pressure in nearby pastoralist areas and insurance coverage in those same areas.

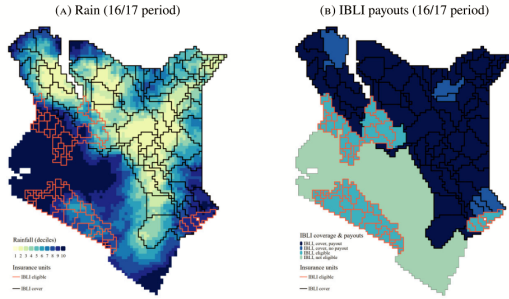


FIGURE III

Droughts (Proxied by Rainfall), IBLI Coverage, and Payouts

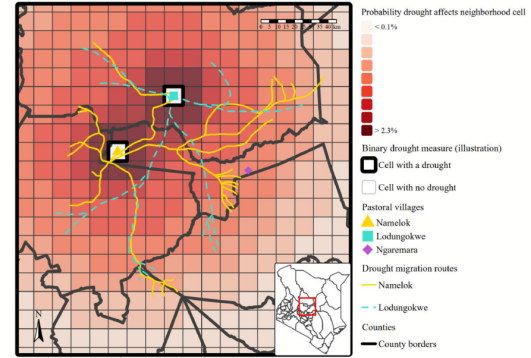


FIGURE IV

Neighborhood Weight Illustration Based on Inverse Distance Weights

6.4 Figure V: Evaluating the Rollout of IBLI + Table I: Baseline Results

Read:

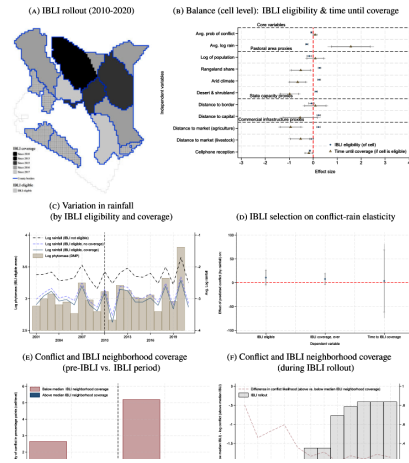
- Figure V documents the stepwise rollout and evaluates whether rollout timing is correlated with pre-existing drought-conflict patterns.
- Panel A shows the rollout pattern over time.
- Panel B reports balance tests for cell and neighborhood characteristics.
- Panel C examines pre-treatment drought-conflict elasticity and rollout selection.
- Table I reports the baseline regression estimates with increasingly rich controls.
- The coefficient on the interaction of neighborhood rain deficit and neighborhood IBLI coverage is negative.
- The implied mitigation is about 25% in the baseline specification.

Detailed interpretation:

- Figure V is an identification figure. It asks whether the rollout appears systematically targeted to places with different pre-trends or different drought-conflict elasticities.
- Table I is the main result table. The relevant coefficient is the interaction, not the level effect of drought or IBLI alone.
- A positive drought coefficient means drought raises conflict risk. A negative interaction means IBLI reduces the marginal effect of drought on conflict.
- Adding controls should not make the coefficient disappear if the effect is not driven by observable rollout selection. The stability of the estimates is therefore important.

Economic message: The main result is not that insured areas are always peaceful. It is that droughts become less conflict-inducing when nearby pastoralist areas have IBLI coverage.

INSURING PEACE: INDEX-BASED LIVESTOCK INSURANCE 1293



INSURING PEACE: INDEX-BASED LIVESTOCK INSURANCE 1297

TABLE I
BASELINE RESULTS

	Dependent variable: $\text{Conflict}_{i,t}$			
	(1)	(2)	(3)	(4)
Neighborhood				
Rain deficit (δ_1)	0.0708 (0.0335)		0.0611 (0.0297)	0.0579 (0.0268)
IBLI (δ_2)		-0.0167 (0.0043)	-0.0252 (0.0052)	-0.0136 (0.0069)
Rain deficit $\times$ IBLI (δ_3)			-0.0156 (0.0052)	-0.0096 (0.0057)
Dep. var. mean	0.0246	0.0246	0.0246	0.0246
Conflict mitigation (%)	—	—	-25.48	-16.66
Cell-level controls	✓	✓	✓	✓
Cell fixed effects	✓	✓	✓	✓
Time fixed effects	✓	✓	✓	✓
Macro-cell time trends	—	—	—	—
Observations	93,400	93,400	93,400	93,400

Notes: The table reports the results of regressing the probability of conflict at the cell level on the rain deficit ($\log(\text{rainfall}) \times -1$), the standardized Index-Based Livestock Insurance (IBLI) coverage, and their respective interaction at both the cell and neighborhood level. Conflict mitigation values in percent are the reduction in the semi-elasticity of the rain deficit on the probability of conflict for a standard deviation increase in the neighborhood IBLI coverage ($\frac{\delta_3}{\delta_1}$). The neighborhood variables are based on the $\frac{1}{\text{distance}}$ weighting scheme. Macro-cell time trends are linear trends on $1^\circ \times 1^\circ$ grid cells, in which our $0.1^\circ \times 0.1^\circ$ cells are nested. If fewer than 55 cells are in a macro cell (5% of the sample at the national borders) we group them into a common border-specific macro cell. Conley standard errors (in parentheses) are implemented using the `acreg` package.

6.5 Figure VI: Dynamic Effects, Placebo Checks, and Randomization Inference + Table II: HSNP and Aid Controls + Figure VII: Overview of Sensitivity Tests

Read:

- Figure VI examines dynamic effects and placebo timing.
- Placebo tests check whether the actual rollout appears to mitigate conflict before coverage exists.
- Randomization inference compares the observed coefficient to coefficients from placebo rollouts.
- Table II controls for other interventions such as HSNP and aid projects.
- Figure VII perturbs the main specification along many dimensions: drought proxies, conflict outcomes, time windows, neighborhood definitions, standard errors, and cell sizes.
- Across these perturbations, the interaction remains negative and generally stable.

Detailed interpretation:

- Figure VI addresses two major concerns: pre-trends and chance findings from spatially correlated data.
- Table II addresses omitted social-protection or aid policies that might independently reduce drought-conflict sensitivity.

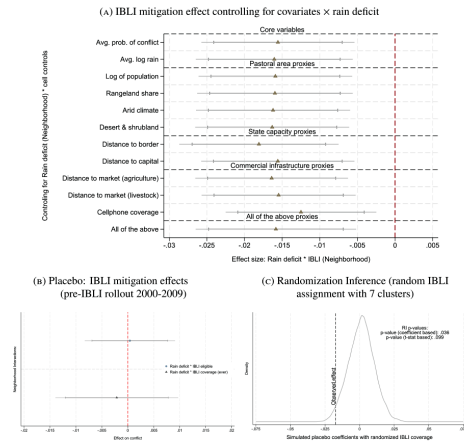


FIGURE VI

Dynamic Effect of Covariates and Randomization Inference

Panel A plots our interaction coefficient of interest δ_3 based on our main specification (Table I, column (3)). In addition, we add the cell-level controls shown as potentially correlated with IBLI coverage (see Figure V, Panel B) and interact them with our neighborhood rain deficit measure. Panel B reports the results from a placebo regression covering the period 2000 to 2009 (based on equation (1) in which we use IBLI eligibility (cell located in IBLI-eligible areas) or IBLI coverage, ever (cell located in IBLI-eligible area), instead of the actual time-varying IBLI coverage. The estimation equation is $(\text{Conflict}_{i,t} = \delta_1 \text{Rain deficit neighborhood}_{i,t} + \delta_2 \text{IBLI neighborhood}_{i,t} + \delta_3 (\text{Rain deficit neighborhood}_{i,t} \times \text{IBLI neighborhood}_{i,t}) + \mathbf{X}_{i,t} \xi + \eta_i + \gamma_t + \epsilon_{i,t})$. Panel C plots the distribution of our neighborhood interaction term between the rain deficit and IBLI cover based on 1,000 randomized IBLI coverage across the rollout clusters. The vertical line represents the estimate we obtain from the observational data. The x-axis is the exact p-value denoting the share of

TABLE II
CONTROLLING FOR OTHER INTERVENTIONS

Other intervention:	Dependent variable: Conflict _{i,t}				
	HSNP		Aid		
	(1)	(2)	(3)	(4)	(5)
Neighborhood					
Rain deficit (δ_1)	0.0611 (0.0297)	0.0600 (0.0302)	0.0508 (0.0318)	0.0710 (0.0303)	0.0844 (0.0304)
IBLI (δ_2)	-0.0252 (0.0052)	-0.0237 (0.0068)	-0.0252 (0.0069)	-0.0264 (0.0053)	-0.0260 (0.0052)
Rain deficit $\times$ IBLI (δ_3)	-0.0156 (0.0052)	-0.0154 (0.0054)	-0.0232 (0.0072)	-0.0179 (0.0053)	-0.0164 (0.0053)
Neighborhood intervention controls					
Intervention		-0.0021 (0.0043)	0.0011 (0.0046)	0.0076 (0.0039)	0.0153 (0.0057)
Rain deficit $\times$ intervention			0.0113 (0.0062)		0.0112 (0.0065)
Dep. var. mean	0.0245	0.0245	0.0245	0.0245	0.0245
Conflict mitigation (%)	-25.48	-25.74	-45.68	-25.16	-19.39
Cell-level controls	✓	✓	✓	✓	✓
Cell fixed effects	✓	✓	✓	✓	✓
Time fixed effects	✓	✓	✓	✓	✓

- Figure VII is the paper's robustness dashboard. It asks whether the headline result is an artifact of one coding decision.
- Robustness is especially important because conflict data are noisy, rainfall-to-forage mapping is imperfect, and migration spillovers require researcher choices.

Economic message: The estimated mitigation effect is not fragile. It survives alternative definitions of drought, conflict, space, time, inference, and overlapping policy controls.

TABLE II
CONTROLLING FOR OTHER INTERVENTIONS

Other intervention:	Dependent variable: Conflict _{i,t}				
	HSNP		Aid		
	(1)	(2)	(3)	(4)	(5)
Neighborhood					
Rain deficit (δ_1)	0.0611 (0.0297)	0.0600 (0.0302)	0.0508 (0.0318)	0.0710 (0.0303)	0.0844 (0.0304)
IBLI (δ_2)	-0.0252 (0.0052)	-0.0237 (0.0068)	-0.0252 (0.0069)	-0.0264 (0.0053)	-0.0260 (0.0052)
Rain deficit $\times$ IBLI (δ_3)	-0.0156 (0.0052)	-0.0154 (0.0054)	-0.0232 (0.0072)	-0.0179 (0.0053)	-0.0164 (0.0053)
Neighborhood intervention controls					
Intervention		-0.0021 (0.0043)	0.0011 (0.0046)	0.0076 (0.0039)	0.0153 (0.0057)
Rain deficit $\times$ intervention			0.0113 (0.0062)		0.0112 (0.0065)
Dep. var. mean	0.0245	0.0245	0.0245	0.0245	0.0245
Conflict mitigation (%)	-25.48	-25.74	-45.68	-25.16	-19.39
Cell-level controls	✓	✓	✓	✓	✓
Cell fixed effects	✓	✓	✓	✓	✓
Time fixed effects	✓	✓	✓	✓	✓

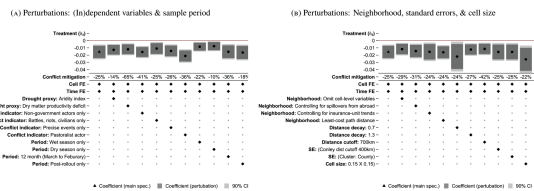


FIGURE VII

Overview: Sensitivity Tests

The figure reports our coefficient of interest (δ_3) over perturbations of our main specification (Table I, column (3)) depicted as a triangle. We introduce alternative drought proxies and conflict measures, add our balancing variables at the cell and neighborhood level, and employ alternative distance-decay functions and standard errors. If not indicated differently, the drought proxy is the rain deficit, and 90% and 95% confidence intervals (in dark and bright gray) are based on Cooley standard errors with a distance cutoff of 200 km. The aridity index is the ratio of precipitation over potential evapotranspiration. Dry matter productivity (DMP) is a phenome indicator measured by the dry biomass increase of the vegetation (in kg/ha/year). Alternative conflict indicators refer to a subset of ACLED events that is used to construct the indicator variable, either based on actors involved, type of events, the geographic position category provided by ACLED, or association of actors with poststates. The periods refer to alternative period constructions in which we use the rain deficit and conflict measures in the month of the dry or wet season only, use 12-month periods from March to February the following year, or drop all periods prior to the IBLI rollout. Controlling for spillovers from abroad controls for the neighborhood rain deficit and neighborhood IBLI coverage from cells within 700 km of Kenya's national border. Insurance unit trends indicate the inclusion of a linear time trend for the 141 insurance units as well as the ineligible area. Least-cost path refers to neighborhood measures in which the inverse distance is based on a least-cost path based on Teller's off-path hiking function (Teller 1980) instead of geographic distance. Distance decays of 0.7 and 1.2 represent flatter and steeper distance decays in the construction of the neighborhood. The cutoff drops all cells beyond 700 km in the construction of the neighborhood measures. The alternative standard errors use a higher distance cutoff when calculating Cooley standard errors or cluster at the county level. See Online Appendix A for details on the variable construction and sources.

6.6 Figure VIII: Extended East Africa Sample + Figure IX: Mixed Settlement and Land-Use Heterogeneity + Table III: Within-Rollout-Cluster Results

Read:

- Figure VIII extends the analysis beyond Kenya to neighboring countries.
- The East Africa extension expands the set of locations not plausibly treated by Kenyan IBLI.
- Figure IX defines contested land-use and mixed settlement areas where pastoralist-farmer interactions are more likely.
- Table III estimates triple-interaction specifications that compare treatment intensity within rollout clusters.
- The mitigation effect is stronger in places where the mechanism predicts more drought-induced conflict: mixed settlement and contested land-use areas.

Detailed interpretation:

- The East Africa extension helps address the concern that Kenya-only trends are driving the result.
- The mixed-settlement and land-use tests are more mechanism-based than simple robustness checks.
- Table III is especially important because rollout-cluster-period fixed effects absorb broad rollout-selection shocks.
- If the effect remains in contested cells within the same rollout cluster and period, it is harder to attribute the result to endogenous rollout timing.

Economic message: IBLI matters most where the economic mechanism is most intense: areas where drought migration creates encounters between pastoralists and other land users.

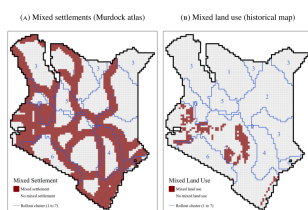
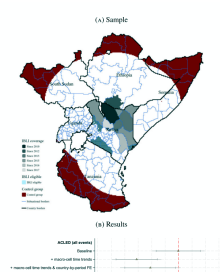


FIGURE IX
Contested Land-Use Classifications

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Contested land-use proxies:	Dependent variable: Conflict _{it}		
	Main: Murdock ethnographic atlas	Alternative: Historical land-use map	
	(1)	(2)	(3)
Neighborhood			
Rain deficit (δ_1)	0.0587 (0.0299)	0.0867 (0.0338)	0.0899 (0.0339)
Rain deficit $\times$ contested land use (δ_2)	0.0039 (0.0105)	0.0057 (0.0095)	-0.0194 (0.0380)
IBLI (δ_3)	-0.0251 (0.0053)	-0.0146 (0.0060)	-0.0145 (0.0067)
IBLI $\times$ contested land use (δ_4)	-0.0030 (0.0031)	-0.0048 (0.0053)	-0.0223 (0.0086)
Rain deficit $\times$ IBLI (δ_5)	-0.0122 (0.0051)	-0.0058 (0.0057)	-0.0097 (0.0059)
Neighborhood mitigation heterogeneity			
Rain deficit $\times$ IBLI $\times$ contested land use (δ_6)	-0.0095 (0.0043)	-0.0114 (0.0044)	-0.0274 (0.0141)
Dep. var. (noncontested)	0.0186 (0.0014)	0.0186 (0.0014)	0.0226 (0.0014)

6.7 Figures X–XIII: Mechanism Evidence

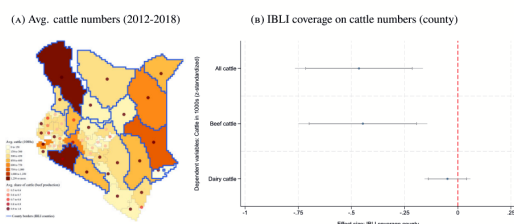


FIGURE X
Herd Size

Panel A plots the average number of cattle in millions across counties for the 2012–2018 period in shades of brown. It also depicts the share of cattle kept for beef production on all cattle. County borders of counties eligible for IBLI are highlighted in blue. The data is obtained from the Kenyan Ministry of Agriculture. Panel B reports the coefficient, 90%, and 95% confidence intervals from regressing the number of cattle z-standardized (for all cattle, cattle kept for beef, and dairy production) on county-averaged IBLI coverage. All specifications include county

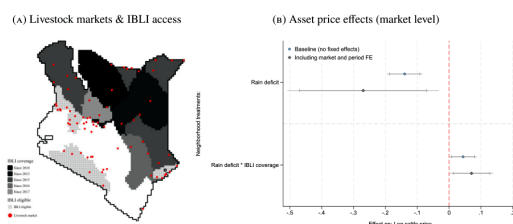


FIGURE XI
Stabilizing Returns to Assets

Panel A depicts livestock markets (red dots) and, in shades of black to light gray, the timing of the IBLI rollout. Panel B depicts the coefficients, 90%, and 95% confidence intervals from regressing the log cattle price on the market level of the two. Dots represent results from a regression without market and period fixed effects, and diamonds represent results from a regression including both period and market fixed effects. Full results, including results on other livestock prices,

Read:

- Figure X studies herd size and shows evidence that IBLI coverage is associated with smaller cattle herds.
- Figure XI studies livestock markets and shows that IBLI mitigates drought-induced cattle price declines.
- Figure XII studies survey-based welfare outcomes such as hunger and no cash income.
- Figure XIII studies the distance between conflict events and ethnic homelands.
- Across these figures, the evidence points to reduced resource pressure, smoother income, more stable asset prices, and less long-distance migration.

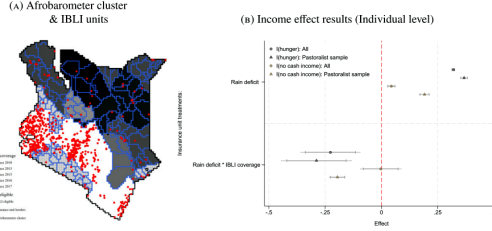


FIGURE XII

Smoothing Adverse Income Shocks

Panel A depicts the Afrobarometer survey cluster (red dots), the insurance unit borders (blue lines), and in shades of black to bright gray, the timing of the IBLI rollout. Panel B depicts the results from regressing the probability of going hungry on the cell-level rain deficit and its interaction with cell-level IBLI coverage, averaged across the 12 months before the respective Afrobarometer survey wave w . The dependent variables are our hunger (black dots and triangles) and no cash income indicators (olive/light-gray dots and triangles). Dots represent results from a regression using all survey respondents. Triangles represent results from restricting the sample survey respondents in the regression to those that identify with an ethnic group in which pastoralism is a common practice (the Borana, Gabra, Kalenjin, Kisumu, Rendile, Mursi, Samak, Somali, Subek, Turkana, and Wabato).

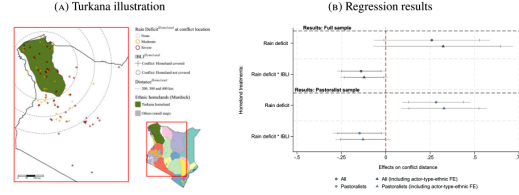


FIGURE XIII

Droughts, IBLI, and Conflict Distance to Ethnic Homelands in Kenya

Panel A plots the conflict locations (ACLED, [Raleigh, Kishi, and Linke 2023](#)) involving Turkana pastoralists over our sample. Different colors indicate the severity of the rain deficit ($\log(\text{rainfall}) \times -1$) in the Turkana homeland (highlighted in solid green). Blue (light-gray) icons refer to years during abundant rainfall in the Turkana homeland, orange (medium-gray) icons refer to locations during moderate droughts in the Turkana homelands, and red (dark-gray) icons indicate conflict locations involving Turkana during severe droughts in their homeland. Moreover, the icon type (+ and o) indicates whether the Turkana homeland was covered by IBLI at the time of the conflict event. The Turkana homeland has been covered by IBLI from 2015 onward (homeland area covered at 98% by IBLI since 2015, see [Figure V](#), Panel A). The rings show the 200, 300, and 400 km distance from the Turkana homeland centroid. The small map in Panel A depicts the different Mur...

Detailed interpretation:

- Figure X supports the rapacity channel: smaller herds reduce pressure on scarce forage and water during drought.
- Figure XI supports a general-equilibrium market channel: fewer distress sales can stabilize cattle prices.
- Figure XII supports an opportunity-cost channel: households face less severe welfare stress during droughts when insurance coverage is higher.
- Figure XIII is direct evidence on migratory pressure: drought usually pushes conflict farther from homelands, and IBLI weakens this effect.
- The mechanisms are complementary. IBLI can reduce conflict through several channels simultaneously rather than through only one narrow effect.

Economic message: IBLI reduces both the incentive and the need for conflict: it lowers pressure on shared resources, cushions household welfare shocks, stabilizes asset values, and reduces risky migration.

7 Detailed result map

7.1 Baseline result block

- Drought in nearby pastoralist regions raises conflict risk.
- IBLI coverage in those same neighborhoods attenuates that effect.
- The main coefficient is the drought $\times$ IBLI interaction.

- The paper interprets the estimate as roughly a 25% reduction in the drought-conflict semi-elasticity for a one-standard-deviation increase in IBLI coverage.

Interpretation. This result is consistent with a conflict-prevention effect, not just a welfare effect. The policy does not need to eliminate drought losses; it only needs to reduce the specific margins that turn drought into violence.

7.2 Robustness result block

- The result is stable across drought proxies and conflict definitions.
- The result is stable when controlling for HSNP and aid.
- Placebo timing does not reveal a pre-existing mitigation effect.
- Randomization inference supports the unusual size of the actual estimate.
- Spatial and temporal perturbations do not overturn the conclusion.

Interpretation. Robustness matters more here than in a simple randomized experiment because rollout is quasi-experimental. The paper builds credibility by showing the pattern is not sensitive to obvious modeling choices.

7.3 Heterogeneity result block

- The effect is stronger where pastoralists are more likely to encounter other land users.
- Mixed settlement and mixed land-use areas are the key high-risk zones.
- Within-rollout-cluster results reduce concerns that broader rollout timing drives the effect.

Interpretation. This is the most mechanism-targeted evidence. If IBLI only reduced general violence, the effect would not necessarily be concentrated in contested land-use areas. The heterogeneity lines up with the migration-conflict story.

7.4 Mechanism result block

- Herd-size reductions suggest insurance substitutes for livestock as precautionary savings.
- Price stabilization suggests fewer simultaneous distress sales.
- Income smoothing suggests a higher opportunity cost of conflict during droughts.
- Shorter conflict distance from homelands suggests lower long-distance migration pressure.

Interpretation. The paper does not observe every individual migration decision, but the mechanisms line up across independent datasets. That makes the overall argument more persuasive.

8 Short summary

- The paper studies whether IBLI mitigates drought-induced conflict in Kenya.
- Droughts create conflict by intensifying competition over grazing resources and pushing pastoralists into contested land-use areas.
- IBLI uses remote-sensing drought indexes to trigger automated, preemptive payouts.
- The main empirical design uses a $0.1^\circ \times 0.1^\circ$ grid-cell panel over 2000–2020.

- The main outcome is an ACLED conflict indicator.
- The main treatment is the interaction between neighborhood drought severity and neighborhood IBLI coverage.
- Neighborhood variables are central because conflict often occurs in migration destinations rather than drought-origin areas.
- A one-standard-deviation increase in neighborhood IBLI coverage reduces the drought-conflict semi-elasticity by about 25%.
- The paper addresses rollout-selection concerns using balance checks, pre-trend checks, placebo timing, randomization inference, alternative samples, and within-rollout-cluster heterogeneity.
- The result remains robust after controlling for HSNP and aid.
- Robustness checks vary drought proxies, conflict definitions, periods, cell sizes, neighborhood decay functions, and inference methods.
- East Africa extensions provide an expanded control set and support the main pattern.
- Contested land-use and mixed settlement heterogeneity support the migration-conflict mechanism.
- Mechanism evidence shows smaller herds, stabilized cattle prices, smoother income, and lower long-distance migration pressure.
- The policy implication is that remote-sensing insurance can be a scalable complement to state-building and peacekeeping in climate-exposed regions.

9 Conclusion

9.1 Core conclusion

Gehring and Schaudt provide quasi-experimental evidence that Index-Based Livestock Insurance can significantly mitigate drought-induced conflict. The main estimate indicates that higher neighborhood IBLI coverage reduces the drought-conflict elasticity by about 25%. This is a large effect because the intervention is not a military, police, or peacekeeping intervention; it is a climate-risk financial product.

9.2 Economic intuition

The economic mechanism is that drought creates scarcity and liquidity stress. Pastoralists face declining forage, pressure to sell livestock, lower cattle prices, hunger, cash shortages, and incentives to migrate farther into areas with other land users. Each of these margins can increase conflict risk. IBLI pays when drought conditions cross a threshold, allowing households to protect livestock, reduce distress sales, smooth income, and avoid some long-distance migration. The conflict-reduction effect emerges because the insurance weakens the channels that turn drought into violence.

9.3 Why the evidence is persuasive

The evidence is persuasive because the paper does not rely on a single result. It combines:

- a spatial case study showing how drought migration generates conflict risk;
- a panel design with rainfall shocks and staggered rollout;
- a negative baseline drought $\times$ IBLI interaction;
- rollout balance and placebo tests;
- controls for HSNP and aid;

- robustness to alternative drought, conflict, spatial, temporal, and inference choices;
- East Africa extensions;
- contested-land heterogeneity; and
- mechanism evidence from herds, prices, welfare, and migration distance.

Taken together, the results are difficult to explain with a single omitted variable.

9.4 Policy implications

- **Climate adaptation.** IBLI is a climate-adaptation tool because it helps households handle drought risk before losses become catastrophic.
- **Conflict prevention.** IBLI can reduce conflict by reducing migration pressure and resource competition.
- **Scalability.** Remote-sensing and mobile payments make the product scalable in low-capacity environments.
- **Complementarity with institutions.** IBLI does not replace political reforms, land governance, or peacekeeping, but it can complement them by reducing the economic shocks that trigger conflict.
- **Targeting.** The largest peace returns may come from coverage in pastoralist origin areas connected to contested destination regions.
- **Subsidies.** Because insurance has public-good conflict-reduction benefits, well-designed subsidies may be justified even if private willingness to pay is limited.

9.5 Limitations

- The rollout is not randomized, so identification depends on rainfall exogeneity, fixed effects, rollout checks, and robustness tests.
- The paper measures coverage and neighborhood exposure, not every household’s actual insurance take-up and usage decision.
- ACLED conflict data may miss some local events or misclassify actors.
- Mechanism tests use different datasets and spatial units, so they are supportive rather than a single unified structural test.
- Scaling may create general-equilibrium effects not fully captured by the historical rollout.
- Future programs such as DRIVE could change prices, take-up, and migration patterns in ways that differ from the historical pilot and rollout period.

9.6 Future research directions

- Measure household-level take-up and payout use more directly.
- Study whether subsidies or information campaigns can increase take-up among the most conflict-relevant households.
- Examine whether large-scale coverage changes livestock prices, migration networks, and land-use negotiations in equilibrium.
- Compare IBLI to cash transfers, public works, or early warning systems as alternative conflict-mitigation tools.
- Study whether insurance products can be bundled with migration-route information, market information, or local peace agreements.

9.7 Final takeaway

The paper's core takeaway is that insurance can be a peace technology. In pastoral economies, drought is not only an income shock; it is a shock to migration, resource competition, asset markets, and social stability. A well-designed insurance product can reduce the probability that climate stress escalates into violence. This makes IBLI an important example of how financial innovation can support both development and peace.